

Python Fundamentals

by Jane Smith

Chapter 1

Python is a high-level interpreted programming language with dynamic typing. It is widely used for scripting, data analysis, and backend web services.

```
def greet(name):  
    print(name)  
    return name
```

Chapter 2

Variables and data types in Python do not require explicit type declarations.

```
x = 42  
name = "Alice"  
items = [1, 2, 3]  
print(type(x))
```

Chapter 3

Functions are first-class objects in Python, enabling functional patterns.

```
import math  
def calculate_area(radius):  
    return math.pi * radius ** 2  
result = calculate_area(5)  
print(result)
```